

VirtualPaint-Tox: label-free virtual Cell Painting with pixel-wise uncertainty for non-destructive toxicity readouts in a liver microphysiological system

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Abstract

[Fill from final results.] Cell Painting, the five-dye morphological profiling assay, is the workhorse of image-based toxicity screening, but staining is terminal: a stained organ-on-chip or microphysiological system (MPS) cannot be followed further. We present VirtualPaint-Tox, a U-Net that predicts all five Cell Painting channels (DNA, ER, RNA, AGP, mitochondria) from a single brightfield image of a liver MPS (OASIS HepatoPAC primary human hepatocyte co-culture), together with a calibrated per-pixel uncertainty map. On held-out wells the virtual stain reaches Pearson r of [x] (DNA) to [y] (AGP), the uncertainty ranks errors with Spearman [z], nuclear counts from the virtual DNA channel agree with the real ones ($r=[]$), and simple morphological profiles from the virtual stain recover treated-vs-control and compound-effect structure [as well as] the real stain. [Axiom U2OS results.] Code, checkpoints and the exact public data subset are released for full reproduction on a laptop.

1 Problem and application scenario

Why it matters. Drug-induced liver injury (DILI) is the most frequent single cause of post-marketing drug withdrawal and a major reason for late-stage attrition, and animal models predict human hepatotoxicity poorly. Human-relevant *in vitro* systems of increasing complexity—from cell lines to primary hepatocytes to micropatterned co-cultures, spheroids and liver-on-chip devices—are therefore being adopted for safety assessment, encouraged by the 3Rs principle and, in the United States, by the FDA Modernization Act 2.0 which removed the animal-testing mandate for investigational drugs. Their richest imaging readout is Cell Painting [6]: five fluorescent dyes label the nucleus, endoplasmic reticulum, nucleoli/RNA, actin/Golgi/plasma membrane and mitochondria, and thousands of morphological features detect bioactivity and cytotoxicity, often at lower concentrations than biochemical viability assays [3]. But the stain is *terminal*: cells are fixed and permeabilised, so a microphysiological system (MPS) that costs hundreds of dollars and is cultured for weeks can be read out exactly once, at one time point, and the fluorescence channels are consumed by the dyes. Brightfield images, in contrast, are acquired on every imager, are free, phototoxicity-free and can be repeated hourly on the same living tissue.

What a label-free virtual stain enables. (i) Longitudinal, non-destructive monitoring of the same chip or well (kinetics of toxicity, recovery, reversibility) instead of one endpoint; (ii) a Cell-Painting-like readout for experiments that cannot afford the dye panel, the extra imaging time or the fixation step; (iii) freeing the fluorescence channels for reporters and antibodies while retaining the morphology profile; (iv) quality control and pre-screening: a cheap brightfield pass flags cytotoxic wells before expensive assays. A virtual stain is only useful, however, if the user knows *where* it is reliable and if the readouts computed from it recover the same biology as the real stain. Both questions are the core of this work.

Sponsor relevance. The challenge sponsor develops neural organ-on-chip platforms and aims at “AI-driven organ-on-a-chip digital twins”. Label-free virtual staining is a building block for such twins: it turns the cheapest, most frequent observation of a chip into a structured, quantitative state description. The method is system-agnostic (we show it on a liver MPS and a cancer cell line and report zero-shot transfer between them) and the training recipe transfers to any bright-field/fluorescence paired data set that a chip laboratory already acquires for validation.

Task. Given a brightfield image x of a field of view, predict the five Cell Painting fluorescence channels $y \in \mathbb{R}^{5 \times H \times W}$ and a per-pixel scale b quantifying the expected error, such that downstream toxicological readouts computed from $\hat{y}$ match those computed from y .

Scope of validation. (i) pixel fidelity per channel on held-out wells; (ii) uncertainty calibration; (iii) biological fidelity: nuclear segmentation, interpretable morphological features, treatment-vs-control separation and compound-effect magnitude; (iv) [\[cytotoxicity \(MTT/LDH\) prediction on Axiom U2OS\]](#).

2 Data, licences and compliance

All data are public, released under CC0 1.0 by the Cell Painting Gallery [1] on the AWS Registry of Open Data (bucket `cellpainting-gallery`); no credentials, paid services or private data are used. No human-subject, clinical or personal data are involved: the biological material is commercially available primary hepatocyte co-culture (HepatoPAC, BioIVT) and the U2OS cell line.

2.1 OASIS HepatoPAC liver MPS (primary dataset)

The OASIS Consortium (HESI and the Broad Institute) [2] profiles a reference set of $\sim 1,500$ compounds with public in-vivo hepatotoxicity data across cell systems of increasing physiological relevance. Its `hepatopac` partition (batch `2026_04_13_OASIS_HepatoPAC_Batch1`) images a micropatterned co-culture of primary human hepatocyte islands and 3T3-J2 stromal cells, a liver MPS used industrially for drug-induced liver injury (DILI) assessment. Acquisition: PerkinElmer Opera Phenix, $20\times$ (NA 1.0), 2×2 binning, 1080×1080 pixels at $0.593 \mu\text{m}/\text{px}$, six channels: brightfield and the five Cell Painting dyes (Hoechst/DNA, concanavalin A/ER, SYTO 14/RNA, WGA+phalloidin/AGP, MitoTracker/Mito). The “Islands” scan covers $48 \text{ wells} \times 49 \text{ fields} = 2,352$ fields (14,112 TIFFs, 33 GB). The platemap lists five anonymised compounds (1–5) at two concentrations (low/high), each in four wells, and eight vehicle (negative-control) wells.

2.2 OASIS Axiom U2OS with biochemical cytotoxicity

[\[describe subset: 50 compounds with full 8-point dose series \(0.046–100 \$\mu\text{M}\$ \) in batch `prod\_25` + all wells with `MTT < 0.7` + 96 DMSO wells; one central field per well; \$2160 \times 2160\$ at \$0.3 \mu\text{m}/\text{px}\$, \$4\times\$ block-averaged to match the HepatoPAC pixel size; per-well MTT and LDH from `biochem.parquet`.\]](#)

2.3 Processing

Fields are stacked to $(6, H, W)$ uint16 arrays after 2×2 block averaging ($1.19 \mu\text{m}/\text{px}$), which quarters compute while keeping nuclei at $\sim 8\text{--}12$ px. Brightfield inputs are standardised per image (median / IQR); fluorescence targets are scaled per channel by the 0.5th/99.8th percentiles of training pixels. **Splits are by well:** 29 training, 7 validation and 12 test wells, stratified by treatment group (every compound/dose and two controls appear in the test set), so no field of a test well is ever seen during training or model selection.

3 Method

3.1 Model

A 2D U-Net (4 resolution levels, 32–512 channels, 7.8 M parameters) maps the brightfield image to five output channels. We add a *heteroscedastic* head: for each channel the network predicts a mean μ and a log-scale $\log b$ of a Laplace distribution and is trained with the negative log-likelihood

$$\mathcal{L} = \frac{1}{N} \sum_{\text{pixels}} \left(\frac{|y - \mu|}{b} + \log b \right) + \lambda_{\text{SSIM}} (1 - \text{SSIM}(\mu, y)), \quad (1)$$

so that b is a calibrated estimate of the expected absolute error and can be displayed as a trust map. Training uses random 256×256 crops with flips/rotations, AdamW (lr $3 \cdot 10^{-4}$, cosine schedule, 200 warm-up steps), batch 16, on Apple-Silicon MPS. Inference runs on full fields with 4-fold flip test-time augmentation (means and scales averaged).

3.2 Baselines and ablations

Per-pixel linear map (no context), a shallow CNN (25 px receptive field, no down-sampling), the same U-Net trained with plain L1 (no uncertainty), with an added SSIM term, and with 25 % / 50 % of the training wells. All ablations use an identical compute budget (1,500 steps); the main model is trained for 5,000 steps and we also report it at the 1,500-step checkpoint for a like-for-like comparison.

3.3 Downstream fidelity protocol

The same label-free pipeline is run on real and virtual channels: (1) Hoechst nuclei are segmented (Gaussian smoothing, Otsu, distance-transform watershed) and counted; objects are matched at $\text{IoU} \geq 0.5$. (2) 49 interpretable field-level features (nuclear count/area/shape; per-channel intensity, nuclear-vs-cytoplasmic ratio, gradient and texture) are computed; we report the per-feature correlation between real and virtual values across test fields. (3) Well profiles (median over fields, standardised to control fields) give a well-by-well similarity matrix (Mantel correlation real vs virtual) and a per-well *effect magnitude* (mean $|z|$ vs controls) whose rank agreement across treated wells is reported. (4) Leave-one-well-out logistic regression separates treated from control fields using real features, virtual features, or brightfield-only texture features (the label-free control that shows what the virtual stain adds).

4 Results

[Tables and figures from results/hepatopac/figures.]

4.1 Pixel fidelity

Table 1: Test-set Pearson correlation per channel (mean over 588 fields of 12 held-out wells, $\pm$ 95% CI over wells).

Model	DNA	ER	RNA	AGP	Mito	mean r	mean SSIM	params (M)
pending								

4.2 Uncertainty calibration

4.3 Biological fidelity

4.4 Cytotoxicity prediction (Axiom U2OS)

4.5 Ablations

5 Reliability analysis and limitations

- **One plate, five anonymised compounds.** The HepatoPAC batch is a single 48-well plate (pre-scan of a larger OASIS campaign); compound identities are anonymised by the data provider. Our well-level split guards against field-level leakage but not against plate-level effects; all treatment-effect conclusions are therefore about *within-plate* biology and need confirmation on further batches as OASIS releases them.
- **Resolution and dimensionality.** Models are trained at $1.19\mu\text{m}/\text{px}$ (2×2 block average) and in 2D, on a single brightfield plane. Thick hepatocyte islands are partly out of focus in both modalities; fine sub-cellular texture (e.g. mitochondrial networks) is beyond what the input carries, which the per-channel results and the uncertainty maps make explicit.
- **Uncertainty is aleatoric.** The Laplace head estimates the irreducible error given the input, not model (epistemic) uncertainty; out-of-distribution inputs (a different cell system, magnification or imager) can be confidently wrong. We quantify this with the cross-system zero-shot transfer experiment and recommend fine-tuning on a small paired set for any new system.
- **Downstream pipeline.** We use a classical nuclear segmentation and 49 hand-designed features so that every number is interpretable and dependency-free; a CellProfiler or deep-feature pipeline would give richer profiles. Conclusions about treated-vs-control separation rest on 12 (HepatoPAC) and 227 (Axiom) held-out wells.
- **Not a regulatory endpoint.** Virtual stains are a monitoring and pre-screening tool; decisions still require the real assay. We report calibration so that users can gate on uncertainty.

6 Impact

[summarise: cost per plate, time, longitudinal designs, 3Rs, digital twins; who uses it (MPS labs, CROs, tox groups); path to deployment (fine-tune on 1 paired plate).]

7 Reproduction

All code is at <https://github.com/datanomadhq/kaggle-ai4s-life-science> (MIT). `python run_all.py --quick` runs every stage on a small subset in about 20 minutes on a laptop; `python run_all.py` downloads the 33 GB HepatoPAC data over plain HTTPS (no account) and reproduces every table and figure; `python run_all.py --dataset axiom` does the same for the U2OS subset. Trained checkpoints (31 MB each) are attached to the GitHub release so that `--stage eval,downstream,figures` regenerates all numbers without training. The exact well-level splits are committed (meta/). Environment: Python 3.12, pinned `requirements.txt`; PyTorch on Apple-Silicon MPS, CUDA or CPU. Training compute: 5,000 steps of batch 16 (2.4 s/step on a shared M-series GPU, ≈ 3.5 h) for the main model, 1,500 steps for each ablation.

8 Third-party resources and licences

- **Data:** Cell Painting Gallery [1], datasets `cpg0037-oasis` (OASIS Consortium [2]: HepatoPAC partition contributed by the Broad Institute imaging platform; Axiom partition by

Axiom Bio), CC0 1.0, hosted by the AWS Open Data Sponsorship Program. We thank the OASIS Consortium, the Broad Institute Imaging Platform and AWS.

- **Software (all OSI licences):** PyTorch (BSD), NumPy, SciPy, pandas, scikit-image, scikit-learn (BSD), tiffle/imagecodecs (BSD), matplotlib (PSF-based), tqdm (MIT/MPL), ffmpeg (LGPL/GPL, video only), macOS say (narration only).
- **Models:** no pretrained weights are used; all networks are trained from scratch in this repository.
- **AI assistance:** code, analysis and text were produced with the help of an AI coding assistant (Claude); all results were generated by the released code on the public data and verified by the team.
- **Prior work:** the method builds on published in-silico labelling [7, 8, 9] and heteroscedastic regression [10]; no prior work of the team is reused.

References

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